

3.1.11 Design for manufacturing, maintenance, repair and disposal

Manufacture, repair, maintenance and disposal

Content	Potential links to maths and science
Students should be aware of, and able to explain, the need to modify designs to make them more efficient to manufacture, including: • reducing the number of manufacturing processes • how the choice of materials affects the use, care and disposal of products: • labelling of materials to aid separation for recycling • making products easy to disassemble or separate • application of the six Rs of sustainability: • reduce the quantity of materials, of toxic materials, of damaging materials and associated energy use • reuse components and parts • rethink by using eco friendly alternative materials • recycle materials and/or components into new products • maintenance: • temporary and integral fixings • use of standardised parts • allowing for service and repair/ replacement of parts • ability to upgrade with software downloads.	

Ease of manufacture

Content	Potential links to maths and science
Students should be aware of, and able to explain, the different ways in which a product can be designed to allow for more efficient manufacture, including: • ribs and webbing to reduce material thicknesses • snap fittings to remove the need for fixings/adhesives • internal moulded screw posts for use with self tapping screws • use of pre made components • use of standardised patterns and sizes • addition of texture in moulding to reduce number of manufacturing processes • self finishing.	

Disassembly

Content	Potential links to maths and science
Students should be aware of, and able to explain, how a product can be designed and manufactured with disassembly in mind, including integral fixings and active disassembly using smart materials such as SMA and biodegradable parts.	